

Question:

Carla is downloading a 200 GB file. Normally she can download 2 GB/minute, but 40% of the way through the download, Windows forces a restart to install updates, which takes 20 minutes. Then Carla has to restart the download from the beginning. How long does it take to download the file?

Reasoning + Answer:

<think> To solve this problem , let's first calculate how much of the file Carla downloaded before the Windows update forced a restart . Since the file size is 200 GB and she has downloaded 40 % of it before the restart , we can find this amount by calculating (200×0.40) . So , $(200 \times 0.40 = 80)$ GB were downloaded before the restart .

Next , we need to account for the time taken during the Windows update , which is 20 minutes . Since we know her download speed is 2 GB per minute , we can determine how much data was left to download after the restart by subtracting the amount she already downloaded (80 GB) from the total file size (200 GB). So , $(200 - 80 = 120)$ GB need to be downloaded after the restart .

Now , to find out the total time taken to download the file , we need to add the time taken before the restart , the time taken for the update (20 minutes), and the time taken after the restart to download the remaining 120 GB . Since she downloads at a rate of 2 GB per minute , we can calculate the time needed to download the remaining 120 GB by dividing 120 by 2 , which gives us $(120 \div 2 = 60)$ minutes .

Finally , we add the time before the restart , the update time , and the time after the restart : $(80 \text{ \textit{ minutes (before restart)}} + 20 \text{ \textit{ minutes (update)}} + 60 \text{ \textit{ minutes (after restart)}})$. Summing these up gives us $(80 + 20 + 60 = 160)$ minutes . </think>

<answer> 160 </answer>

